

WEEK 5 ACCOMPLISHMENT REPORT

CAPSTONE PROJECT TITLE: GeoSentri: An IoT-Based Real-Time GPS Monitoring System for Police Personnel
PERCENTAGE OF COMPLETION: 40-50%
These tasks were developed during Week 5, then consolidated and committed to Git on August 4, 2026. The table shows the verified Git commit date.

DATE	DAY	COMPLETED TASK	PERSON RESPONSIBLE
August 4, 2026	Tuesday	Started the Flespi GPS and MQTT integration for personnel location updates.	All Members
August 4, 2026	Tuesday	Prepared the real-time data flow for personnel, deployments, tasks, and reports.	All Members
August 4, 2026	Tuesday	Prepared the supervisor deployment and assigned-area management workflow.	All Members
August 4, 2026	Tuesday	Prepared the officer task viewing and task-status workflow.	All Members
August 4, 2026	Tuesday	Developed the initial report submission flow with location and photo evidence.	All Members

WEEK 6 ACCOMPLISHMENT REPORT

CAPSTONE PROJECT TITLE: GeoSentri: An IoT-Based Real-Time GPS Monitoring System for Police Personnel
PERCENTAGE OF COMPLETION: 50-55%

DATE	DAY	COMPLETED TASK	PERSON RESPONSIBLE
August 3, 2026	Monday	Completed the integration of real-time operational workflows across backend, web, and mobile.	All Members
August 4, 2026	Tuesday	Completed the web personnel monitoring map and operational dashboard.	All Members
August 4, 2026	Tuesday	Completed the mobile map, reports, tasks, profile, and account workflows.	All Members
August 4, 2026	Tuesday	Added reverse geocoding and GPS-reading freshness tracking.	All Members
August 5, 2026	Wednesday	Updated mobile behavior, Android build configuration, and Expo dependencies.	All Members

WEEK 7 ACCOMPLISHMENT REPORT

CAPSTONE PROJECT TITLE: GeoSentri: An IoT-Based Real-Time GPS Monitoring System for Police Personnel
PERCENTAGE OF COMPLETION:55-65%

DATE	DAY	COMPLETED TASK	PERSON RESPONSIBLE
August 11, 2026	Tuesday	Added shift scheduling notifications and personnel profile photos.	All Members
August 12, 2026	Wednesday	Polished mobile workflows and prepared the application for production builds.	All Members
August 12, 2026	Wednesday	Finalized GeoSentri Android branding, notification behavior, and secondary officer testing support.	All Members
August 13, 2026	Thursday	Implemented native and web MapLibre maps using MapTiler styles.	All Members
August 13, 2026	Thursday	Improved live marker animation, marker clustering, map colors, zoom behavior, and mobile map controls.	All Members
August 14, 2026	Friday	Refined responsive map layouts, deployment views, action styling, authentication UX, security, and performance.	All Members
August 15, 2026	Saturday	Strengthened operational security, upgraded the mobile SDK, and fixed tracking and deployment state synchronization.	All Members
August 16, 2026	Sunday	Fixed mobile evidence uploads, offline report storage, and dark-theme surfaces.	All Members
August 16, 2026	Sunday	Polished mobile map interactions and isolated native map caching from the web preview.	All Members
August 16, 2026	Sunday	Added authentication-session support for the mobile web preview.	All Members

WEEK 8 ACCOMPLISHMENT REPORT

CAPSTONE PROJECT TITLE: GeoSentri: An IoT-Based Real-Time GPS Monitoring System for Police Personnel
PERCENTAGE OF COMPLETION: 65-70%

DATE	DAY	COMPLETED TASK	PERSON RESPONSIBLE
August 19, 2026	Wednesday	Hardened security across the backend, web portal, and mobile application.	All Members
August 19, 2026	Wednesday	Added security headers, authorization checks, and request rate limiting.	All Members
August 19, 2026	Wednesday	Strengthened profile-photo and report-evidence upload validation.	All Members
August 21, 2026	Friday	Fixed real-time operational workflows and refined the web and mobile user experience.	All Members
August 21, 2026	Friday	Added encrypted offline report queuing and automatic synchronization.	All Members
August 22, 2026	Saturday	Improved operational monitoring, deployment management, and supervisor workflows.	All Members
August 22, 2026	Saturday	Added barangay operational analytics and incident visualization.	All Members
August 22, 2026	Saturday	Added an evidence viewer and expanded report-detail presentation.	All Members
August 22, 2026	Saturday	Added map legends, loading skeletons, confirmations, and accessible dialogs.	All Members
August 23, 2026	Sunday	Improved authentication safeguards and user guidance during login.	All Members

WEEK 9 ACCOMPLISHMENT REPORT

CAPSTONE PROJECT TITLE: GeoSentri: An IoT-Based Real-Time GPS Monitoring System for Police Personnel
PERCENTAGE OF COMPLETION: 70-80%

DATE	DAY	COMPLETED TASK	PERSON RESPONSIBLE
August 26, 2026	Wednesday	Polished the light and dark themes of the monitoring dashboard.	All Members
August 27, 2026	Thursday	Improved frontend workflows, page behavior, and navigation usability.	All Members
August 27, 2026	Thursday	Improved account forms, validation, and GPS-device selection.	All Members
August 27, 2026	Thursday	Improved deployment scheduling, personnel selection, and deployment dialogs.	All Members
August 27, 2026	Thursday	Improved report workflows and real-time interface feedback.	All Members
August 28, 2026	Friday	Refactored operational backend domains to improve maintainability and reliability.	All Members
August 28, 2026	Friday	Separated deployment, report, and task logic into maintainable backend services.	All Members
August 28, 2026	Friday	Added a dedicated real-time socket gateway and operational runtime.	All Members
August 28, 2026	Friday	Added report and task pagination with offline synchronization support.	All Members
August 28, 2026	Friday	Added shared domain contracts, automated tests, and continuous-integration checks.	All Members

WEEK 10 ACCOMPLISHMENT REPORT

CAPSTONE PROJECT TITLE: GeoSentri: An IoT-Based Real-Time GPS Monitoring System for Police Personnel
PERCENTAGE OF COMPLETION: 80-90%

DATE	DAY	COMPLETED TASK	PERSON RESPONSIBLE
September 1, 2026	Tuesday	Improved map controls and real-time GPS monitoring reliability.	All Members
September 3, 2026	Thursday	Fixed authentication pagination and mobile connectivity to the AWS-hosted backend.	All Members
September 3, 2026	Thursday	Added safe reconciliation of the supervisor login account.	All Members
September 4, 2026	Friday	Strengthened account email validation and aligned the mobile application with Expo SDK 57.	All Members
September 4, 2026	Friday	Refined mobile report filters, bottom navigation, and preview APK architecture.	All Members
September 4, 2026	Friday	Added project and AWS infrastructure documentation.	All Members
September 6, 2026	Sunday	Improved system reliability and refreshed the personnel-management interface.	All Members
September 6, 2026	Sunday	Improved mobile navigation and responsive screen layouts.	All Members
September 6, 2026	Sunday	Added system health endpoints, protected-route recovery, and page-data caching.	All Members
September 6, 2026	Sunday	Improved automatic recovery and synchronization of offline mobile reports.	All Members

WEEK 11 ACCOMPLISHMENT REPORT

CAPSTONE PROJECT TITLE: GeoSentri: An IoT-Based Real-Time GPS Monitoring System for Police Personnel

PERCENTAGE OF COMPLETION: 90-99%

DATE	DAY	COMPLETED TASK	PERSON RESPONSIBLE
September 7, 2026	Monday	Evaluated GPS-reading age, freshness indicators, and test markers on the mobile map.	All Members
September 7, 2026	Monday	Checked marker legend icons and the spacing of the officer-following panel.	All Members
September 8, 2026	Tuesday	Conducted functional testing of marker movement and clustering using incoming GPS updates.	All Members
September 8, 2026	Tuesday	Tested automatic OTP verification and the visible manual verification fallback.	All Members
September 8, 2026	Tuesday	Reproduced and fixed the keyboard issue after entering an incorrect verification code.	All Members
September 8, 2026	Tuesday	Evaluated service-outage, network, and server error messages across web pages.	All Members
September 9, 2026	Wednesday	Tested report-validation notifications and direct navigation to the exact report, including status badges, reviewer details, and review time.	All Members
September 9, 2026	Wednesday	Evaluated report editing, correction reasons, and revision history, and confirmed corrected validated reports return to pending for repeat validation.	All Members
September 9, 2026	Wednesday	Verified the completeness of report details, automatic fields, evidence, and incident time.	All Members
September 9, 2026	Wednesday	Validated current-GPS checks and account-wide OTP limiting across devices, with automated regression tests for the corrected workflows.	All Members
September 13, 2026	Sunday	Made report, backup request, boundary, inactivity, and deployment notifications open their exact related report, officer, map, or assignment.	All Members
September 13, 2026	Sunday	Added a notification-based map view that focuses on the requesting officer and displays backup request details, location, time, responder count, and completion action.	All Members
September 13, 2026	Sunday	Added the mobile backup-response completion flow, including confirmation, SOS closure, and a two-minute red emergency overlay for the requesting officer.	All Members
September 13, 2026	Sunday	Added immediate or later incident-report creation after a backup response, with direct access to the linked report from task history.	All Members
September 13, 2026	Sunday	Automatically linked completed backup requests to incident reports and recorded the request location, GPS source, assigned area, response team, and responder details.	All Members
September 13, 2026	Sunday	Added validation against premature, unauthorized, or duplicate backup-linked reports and verified the workflow with backend, web, and mobile regression tests.	All Members